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TEB – 103

ETE

a)

Material	project Id	space group
NiO	mp-19009	$Fm\bar{3}m$
CoO	mp-22408	$F\bar{4}3m$
FeO	mp-1274279	$C2/m$
TiO2	mp-390	$I4_1/amd$
RuO2	mp-825	$P4_2/mnm$
IrO2	mp-2723	$P4_2/mnm$
SrTiO3	mp-5229	$Pm\bar{3}m$
LaMnO3	mp-17554	Pnma

b)

To quantitatively isolate and measure the D3-induced error, the following protocol was used:

Step 1: Selection of optimized structure

The optimized crystal structures of all eight transition metal oxides were obtained from the Materials Project database.

Step 2: Uniform lattice expansion

Each crystal structure was uniformly expanded by applying strain to the lattice parameter:

$$a_i = a_0(1 + s_i)$$

where:

- a_0= original lattice parameter
- s_i= applied strain
- a_i= strained lattice parameter

Step 3: D3 dispersion energy calculation

For every strained structure, the D3 dispersion energy (E_{dis})was calculated and recorded.

Step 4: Plotting dispersion-energy curve

A graph of:

$$E_{dis} \ vs \ \text{Cell Length}$$

was plotted for each material.

Step 5: Identification of dip

The abnormal minimum point (dip) in the curve was identified:

$$E_{min}$$

This point corresponds to the strongest artificial lowering of D3 energy.

Step 6: Estimation of smooth baseline

A smooth baseline was constructed by connecting the pre-dip shoulder and post-dip recovered region:

$$E_{expected}(x) = mx + b$$

where:

$$m = \frac{E_R - E_L}{x_R - x_L}$$
$$b = E_L - m x_L$$

Step 7: Calculation of D3-induced error

The D3-induced error was calculated as:

$$\Delta E = E_{min} - E_{expected,dip}$$

where:

- E_{min}= observed minimum dispersion energy
- $E_{expected,dip}$= expected smooth energy at dip position

Magnitude:

$$| \Delta E |$$

represents the strength of D3-induced error.

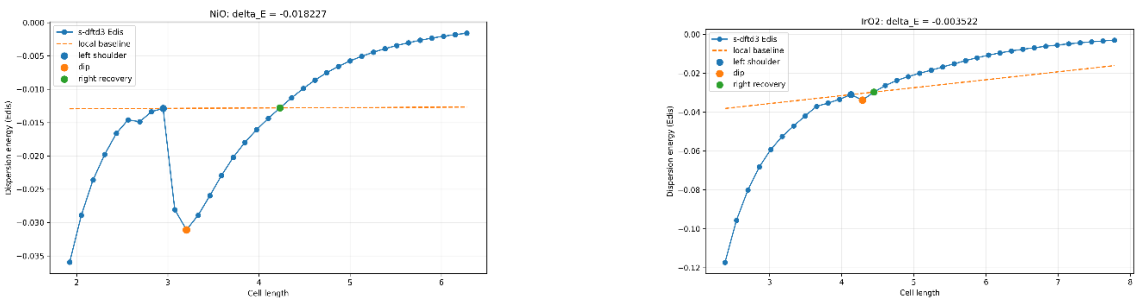

Figure: *Representative dispersion-energy vs cell length plot showing pre-dip shoulder, artificial D3 dip (E_{min}), and recovered post-dip region used for baseline fitting.*

c)

The D3-induced error contribution was extracted for all eight transition metal oxides using the local peak-dip-recovery baseline method. The error was calculated using:

$$\Delta E = E_{\text{min}} - E_{\text{expected,dip}}$$

where E_{min} is the observed minimum dispersion energy at the dip point, and $E_{\text{expected dip}}$ is the expected smooth baseline energy at the same position. The magnitude $|\Delta E|$ represents the D3-induced error contribution.

Material	**Material Project ID**	**Space Group**	**Dip Cell Length**	E_{min}	$E_{expected,dip}$	**ΔE**	$\lvert \Delta E \rvert$
NiO	mp-19009	$Fm\bar{3}m$	3.204498	-0.031086	-0.012859	-0.018227	0.018227
CoO	mp-22408	$F\bar{4}3m$	3.468908	-0.042808	-0.019069	-0.023739	0.023739
FeO	mp-1274279	$C2/m$	2.720810	-0.091038	-0.046713	-0.044324	0.044324
TiO2	mp-390	$I4_1/amd$	5.144189	-0.033192	-0.013372	-0.019820	0.019820
RuO2	mp-825	$P4_2/mnm$	3.422476	-0.047043	-0.038392	-0.008651	0.008651
IrO2	mp-2723	$P4_2/mnm$	4.288553	-0.033895	-0.030374	-0.003522	0.003522
SrTiO3	mp-4651	$Pm\bar{3}m$	7.174065	-0.100512	-0.038759	-0.061753	0.061753
LaMnO3	mp-17554	Pnma	8.513574	-0.222176	-0.089542	-0.132634	0.132634

From the table, **$LaMnO_3$** has the largest D3-induced error contribution, while **IrO_2** has the smallest. The overall order of error magnitude is:

$$LaMnO_3 > SrTiO_3 > FeO > CoO > TiO_2 > NiO > RuO_2 > IrO_2$$

d)

The calculated D3-induced error magnitudes follow the order:

$$LaMnO_3 > SrTiO_3 > FeO > CoO > TiO_2 > NiO > RuO_2 > IrO_2$$

This trend indicates that D3-induced error depends strongly on **structure type, ionic size, oxidation state, packing density, and electronic/geometric distortions**.

Table 1: Summary of D3-induced error trend

Material	Structure Type	$\lvert \Delta E \rvert$	Key Reason for Trend
$LaMnO_3$	Distorted Perovskite	0.132634	Large La ion + Jahn–Teller distortion + open structure
$SrTiO_3$	Perovskite	0.061753	Large Sr ion + extended TiO_6 network
FeO	Distorted Rock Salt	0.044324	Lower symmetry + distorted Fe–O coordination
CoO	Rock Salt	0.023739	Moderate ionic packing
TiO_2	Oxide Network	0.019820	Relatively regular TiO_6 arrangement
NiO	Rock Salt	0.018227	Compact symmetric ionic lattice
RuO_2	Rutile	0.008651	Dense packing + smooth coordination change
IrO_2	Rutile	0.003522	Most compact regular structure

Effect of structure type

Perovskite oxides show the highest D3-induced errors because they contain **multiple coordination environments, large cation sites**, and **extended octahedral frameworks**. During expansion, many weak long-range interactions change simultaneously, producing a stronger artificial dip.

Rock salt oxides show moderate error because their packing is compact and coordination is simpler.

Rutile oxides show the smallest error due to **dense packing** and **regular octahedral arrangement**, which keeps D3 variation smooth.

Table 2: Critical factors affecting D3-induced error

Factor	Effect on D3-induced Error	Example
Large ionic size	Increases polarizability → increases error	La^{3+}, Sr^{2+}
Open crystal packing	More weak long-range interactions	Perovskites
Structural distortion	Causes abrupt coordination change	$LaMnO_3$, FeO
High symmetry	Smooth D3 variation → smaller error	NiO, IrO_2
Dense packing	Reduces artificial dip	Rutile oxides

Effect of ionic size and oxidation state

Large ions such as **La^{3+}** and **Sr^{2+}** have high polarizability, increasing weak dispersion interactions and resulting in larger D3-induced errors.

Higher oxidation state ions such as **Ti^{4+}, Ru^{4+},** and **Ir^{4+}** are smaller and more tightly bound, reducing dispersion-related artifacts.

Electronic and geometric effects

$\boldsymbol{LaMnO_3}$contains **Jahn–Teller active $\boldsymbol{Mn^{3+}}$**, producing unequal Mn–O bond lengths and strong structural distortion, which explains its highest D3-induced error.

$IrO_2$is an anomaly: despite heavy Ir atoms, it shows the lowest error because its **compact rutile geometry dominates over atomic mass effects.**

Conclusion

$$\text{Perovskite} > \text{Rock Salt} > \text{Rutile}$$

Thus**, open, distorted, and highly polarizable structures show the largest D3-induced errors,** whereas **dense, symmetric, and compact structures show the smallest errors.**